

Prometis Bank Loan Approval Automation – Analysis Report

Introduction

This report outlines the exploration of automating loan approval decisions at Prometis Bank using historical loan data. The aim was to build a predictive model that could assist human decision-makers while meeting compliance and transparency requirements stipulated by the EU AI Act. In this analysis, I examine the dataset, develop several predictive models, and evaluate model performance across multiple metrics. Given the need for transparency and interpretability in high-risk financial applications, significant attention is given to model explainability and responsible deployment strategies.

Data Analysis and Feature Engineering

Dataset Overview

The analysis was based on a cleaned dataset of 50,000 loan applications approved between January 2020 and December 2022. The dataset comprises 19 features including borrower demographics, loan characteristics, and the loan outcome. The historical data originates from applications approved by expert loan officers, but it lacks records for rejected applications. This introduces a potential selection bias because the model lacks contrastive examples (rejected applications), which hinders its ability to learn true decision boundaries and can lead to biased risk assessments. Including rejected-application data would enable learning of true decision boundaries between approved and declined cases, reduce sample selection bias, and improve fairness by ensuring that subgroups underrepresented in approved-only data are properly modeled, yielding more reliable approval recommendations in production.

Feature Engineering

In addition to the raw features provided, I created financial and risk-proxy variables to capture borrower capacity and regional/age-based risk patterns. Among other variables, I constructed disposable income (monthly income minus the new loan payment) and the savings-to-loan ratio (savings divided by requested loan amount). I also generated proxies for geographic and demographic risk by mapping each borrower's zip code prefix and age group to the historical default rates of their respective cohorts. This transforms sensitive or high-cardinality categories into continuous risk scores that capture local economic stress and life-stage vulnerability without directly encoding demographic attributes, thereby enhancing predictive power while mitigating unfair bias.

Modeling Approach and Evaluation

Model Development

I implemented and compared only four wide-used models for rapidity in the execution: Logistic Regression (baseline), Decision Tree, Random Forest, and XGBoost. A two-stage feature selection process was used: (1) an initial logistic regression to identify the top 25 features by coefficient significance; (2) manual re-introduction of economically important variables such as loan-to-income ratio and previous defaults. The refined feature set was used to train a further logistic regression (henceforth called **Logistic Regression 2**).

Logistic Regression 2 achieved an accuracy of 0.773 and an AUC of 0.835, showing a marginal improvement over the baseline logistic regression while retaining interpretability. XGBoost

Model	Accuracy	AUC	Precision	Recall	F1 Score
Logistic Regression	0.771	0.835	0.711	0.636	0.671
Logistic Regression 2	0.773	0.835	0.714	0.640	0.675
Decision Tree	0.763	0.820	0.691	0.640	0.665
Random Forest	0.773	0.831	0.715	0.636	0.673
XGBoost	0.774	0.835	0.714	0.642	0.676

Table 1: Model Performance Comparison

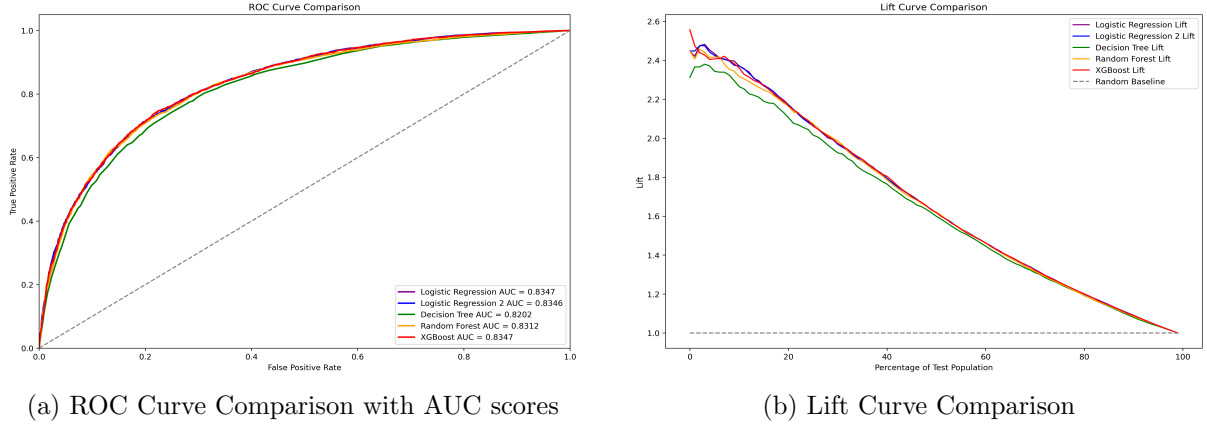


Figure 1: Pair of model evaluation curves

delivered the highest accuracy (0.774), but the performance gain is marginal (0.1 percentage points) and is outweighed by transparency requirements under the EU AI Act, which favor interpretable models like Logistic Regression.

Comparison: Logistic Regression 2 vs. XGBoost

The comparative analysis between Logistic Regression 2 and XGBoost reveals nuanced trade-offs between predictive performance and regulatory compliance. While XGBoost achieves marginally higher accuracy as shown in Table 1, Logistic Regression 2 emerges as the superior choice for deployment in this high-risk financial context.

The ROC curves in Figure 1a demonstrate near-identical AUC values for both models, indicating comparable discriminative power. Similarly, the lift curves in Figure 1b show that both models maintain stable gains over the random baseline across deciles of the test population. However, XGBoost’s ensemble structure obscures decision logic, whereas Logistic Regression 2 provides linear coefficients that directly map features to outcomes.

The EU AI Act mandates that high-risk AI systems be “transparent, traceable, and subject to human oversight.” Logistic Regression 2 satisfies these requirements through its inherently interpretable architecture. Each feature’s contribution to the loan approval decision can be quantified as $\log\text{-odds} = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$, enabling clear explanations to borrowers and regulators. In contrast, XGBoost’s black-box nature complicates compliance on transparency obligations from the EU AI Act. Table 1 further shows that both models achieve similar precision, recall, and F1 scores, negating any material advantage for XGBoost in classification quality.

EU AI Act Considerations

To align with the requirements of the EU AI Act, the model development process incorporated several critical safeguards. A risk management system was implemented to ensure model reliability. Stratified cross-validation assessed generalization across diverse economic conditions, while regularization techniques were employed to mitigate overfitting. However, a recognized limitation lies in the historical dataset’s exclusion of rejected loan applications; addressing this gap by integrating such records remains a priority for future iterations to reduce selection bias and enhance representativeness.

Transparency and documentation formed the backbone of compliance efforts. Comprehensive logs were maintained, detailing feature definitions, model configurations, and SHAP interpretability outputs. These logs ensure full auditability, enabling regulators and internal stakeholders to trace decision logic and validate algorithmic fairness. The logistic regression model enables loan officers to manually override automated decisions based on predefined risk thresholds or exceptional circumstances, aligning with Article 14 of the EU AI Act, which mandates meaningful human oversight for high-risk systems. This balances efficiency with accountability, ensuring high-risk cases receive expert scrutiny.

Non-discrimination and fairness were proactively addressed through proxy-variable analysis, which confirmed the absence of undue biases in demographic or geographic features. The use of engineered risk proxies—such as regional default rates instead of raw zip codes—minimized direct encoding of sensitive attributes. Despite these safeguards, further fairness audits are recommended, particularly through synthetic data generation with protected labels, to stress-test the model under hypothetical bias scenarios and fortify its ethical foundations. Collectively, these strategies ensure compliance with the EU AI Act’s mandates while maintaining operational resilience.

Conclusion

This analysis demonstrates that automated loan approval systems can enhance efficiency at Prometis Bank while adhering to the EU AI Act’s stringent requirements. Logistic Regression 2 is recommended for deployment due to its balance of interpretability and performance. Though XGBoost achieves marginally higher accuracy, its black-box nature conflicts with regulatory transparency obligations. The model’s linear coefficients enable loan officers to explain decisions to borrowers and auditors, fulfilling Article 13 requirements for high-risk systems.

A critical limitation lies in the dataset’s exclusion of rejected applications, which introduces selection bias and restricts the model’s ability to generalize. Future efforts must prioritize collecting data on declined loans to improve fairness and accuracy. Additionally, ongoing monitoring for demographic bias and economic drift is essential to maintain compliance and reliability.

To ensure responsible deployment, Prometis Bank should:

- Integrate rejected-application data to mitigate selection bias
- Implement real-time monitoring for model drift and fairness metrics
- Conduct quarterly audits using synthetic data to test bias scenarios
- Maintain human oversight protocols for high-risk or edge cases

By adopting these measures, Prometis Bank can leverage automation responsibly, aligning operational efficiency with regulatory compliance and ethical standards.